RAGUL PAZHAMALAI

Applied AI Engineer | Generative AI | RAG | Python

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PROFESSIONAL SUMMARY

AI-focused engineer with 4+ years of enterprise analytics and automation experience, combining hands-on Generative AI and machine learning projects with Power BI and Power Platform expertise. Built an advanced PDF RAG application with hybrid retrieval, multi-query search, contextual compression, conversational memory, evaluation and a Streamlit interface. Microsoft PL-300 certified, with experience translating business requirements into analytical applications and secure enterprise reporting solutions.

TECHNICAL SKILLS

Generative AI: RAG, LangChain, OpenAI API, prompt engineering, embeddings, Chroma, semantic search, BM25, parent-child retrieval, multi-query retrieval, Reciprocal Rank Fusion (RRF), contextual compression, conversational memory

Python & Machine Learning: Python, SQL, pandas, NumPy, scikit-learn, data preprocessing, classification, regression, model evaluation

Applications & Evaluation: Streamlit, PyPDFLoader, Git/GitHub, groundedness, context relevance, Hit@K, Mean Reciprocal Rank (MRR)

Power Platform & Analytics: Power BI, Power BI Service, Power Apps, Power Automate, DAX, Power Query, ETL, data modeling, Row-Level Security (RLS), Excel

FEATURED AI PROJECT

Advanced PDF Question Answering System | RAG

Personal project | Python, LangChain, OpenAI API, Chroma, BM25, Streamlit

GitHub: github.com/ragulpazhamalai/pdf-rag-chatbot

- Built an end-to-end PDF question answering pipeline using document loading, chunking, OpenAI embeddings, Chroma vector storage and LLM response generation.
- Implemented parent-child retrieval to retrieve focused chunks while supplying broader source context to the language model.
- Combined BM25 keyword search with vector similarity search; used multi-query generation and Reciprocal Rank Fusion to combine ranked retrieval results.
- Added contextual compression to focus the supplied context and conversational memory to support follow-up questions across turns.
- Added document-grounded answer instructions and fallback handling for questions unsupported by the source document.
- Built an evaluation workflow covering groundedness, context relevance, Hit@K and MRR, with retrieval debugging to inspect search behavior.
- Created a Streamlit chat interface for interacting with the RAG system.

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Applied AI Engineer | Enterprise Analytics & Power Platform

PROFESSIONAL EXPERIENCE

Senior Data Analyst | Randstad

September 2023 - Present | Chennai | Client: Caterpillar

- Managed and maintained 180+ Power BI reports, supporting performance optimization, timely enhancements and report availability.
- Developed interactive dashboards and KPI reports; designed data models using DAX and Power Query to support business decision-making.
- Performed ETL using SQL and Power Query to prepare clean, reliable datasets for enterprise reporting.
- Partnered with business stakeholders to gather requirements, deliver analytical insights and support reporting governance, security and performance improvements.

Data Analyst | Onward Technologies

July 2022 - September 2023 | Chennai | Client: Caterpillar

- Built Power Apps solutions to digitize manual business processes and improve operational efficiency.
- Developed Leave/WFH Management and ECR Request applications with approval workflows.
- Used SQL and Excel to analyze data, identify trends and generate business insights; collaborated with cross-functional teams to automate reporting and improve process efficiency.

ENTERPRISE ANALYTICS PROJECT

Drawing Quality & Root Cause Analysis Dashboard

Power BI | DAX | Power Query | Data Modeling | RLS

- Developed an end-to-end dashboard covering the engineering drawing lifecycle from detailing to manufacturing release, serving 1,500+ users across multiple teams.
- Designed a snowflake schema with fact and dimension tables; developed cycle time, process velocity and error rate KPIs to track engineering performance.
- Used root cause analysis and conditional formatting to surface bottlenecks and projects exceeding target turnaround time; helped reduce drawing release delays by approximately 15-20%.
- Configured Row-Level Security with four access roles to provide secure project-level visibility.

CERTIFICATION

Microsoft Certified: Power BI Data Analyst Associate (PL-300)

EDUCATION

Bachelor of Engineering - Mechanical Engineering

SA Engineering College, Chennai | 2014 - 2018